

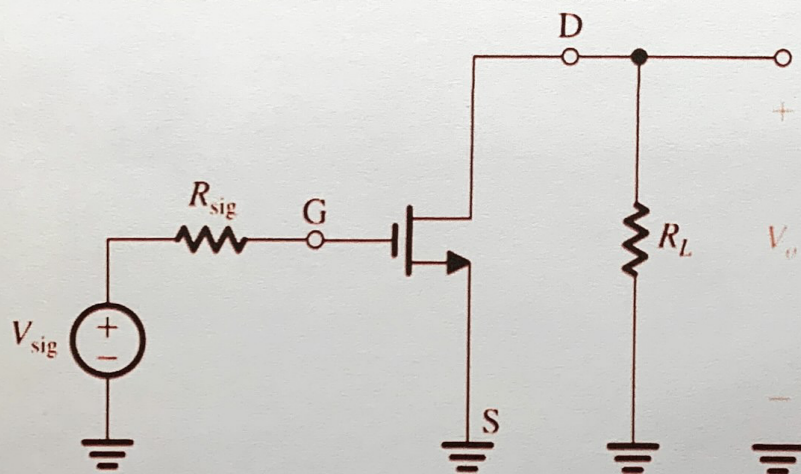
作业

10.6 For the CS amplifier specified in Example 10.1, find the values of A_M and f_H that result when the load resistance is reduced to $10\text{ k}\Omega$.

Ans. -13.3 V/V ; 86.9 MHz

Example 10.1

Find the midband gain A_M and the upper 3-dB frequency f_H of an integrated-circuit CS amplifier fed with a signal source having an internal resistance $R_{\text{sig}} = 20\text{ k}\Omega$. The amplifier has $R_L = 20\text{ k}\Omega$, $g_m = 2\text{ mA/V}$, $r_o = 20\text{ k}\Omega$, $C_{gs} = 20\text{ fF}$, and $C_{gd} = 5\text{ fF}$. Also, find the frequency of the transmission zero.



(a)

$$A_M = -g_m(R_L \parallel r_o) = -13.33\text{ V/V}$$

$$C_{in} = C_{gs} + (1 - A_M)C_{gd} = 91.65\text{ fF}$$

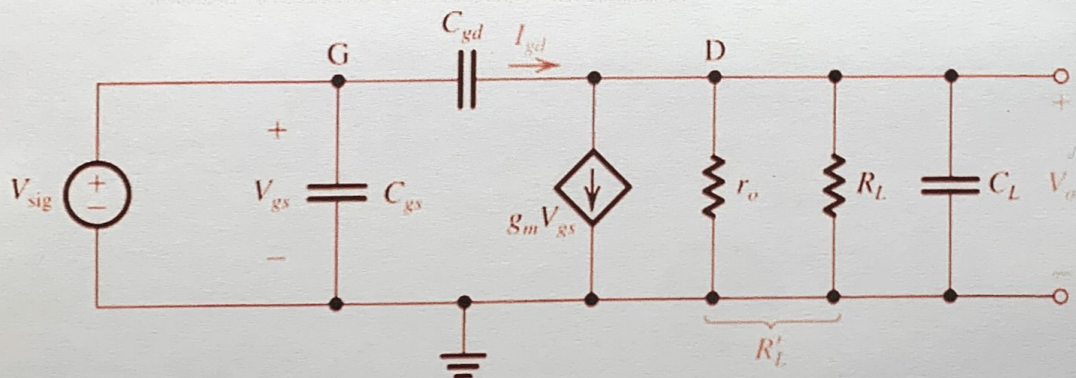
$$f_H = \frac{1}{2\pi C_{in} R_{\text{sig}}} = 86.87\text{ MHz}$$

10.8 For the CS amplifier considered in Example 10.2 operating at the original values of V_{ov} and I_D , find the value to which C_L should be increased to place f_t at 3 GHz.

Ans. 101 fF

Example 10.2

Consider an IC CS amplifier fed with a source having $R_{sig} = 0$ and having an effective load resistance R'_L composed of r_o of the amplifier transistor in parallel with an equal resistance r_o of the current-source load. Let $g_m = 2 \text{ mA/V}$, $r_o = 20 \text{ k}\Omega$, $C_{gs} = 20 \text{ fF}$, $C_{gd} = 5 \text{ fF}$, and $C_L = 25 \text{ fF}$. Find A_M , f_H , f_t , and f_z . If the amplifying transistor is to be operated at twice the original overdrive voltage while W and L remain unchanged, by what factor must the bias current be changed? What are the new values of A_M , f_H , f_t , and f_z ?



$$f_t = |A_M| \cdot f_H$$

$$f_{t1} = \frac{1}{2\pi(C_L + C_{gd}) \cdot R'_L} = 10.6 \text{ GHz} \quad \dots \textcircled{1}$$

(见教材 7.694)

$$f_{t2} = \frac{1}{2\pi(C_L^* + C_{gd}) \cdot R'_L} = 3 \text{ GHz} \quad \dots \textcircled{2}$$

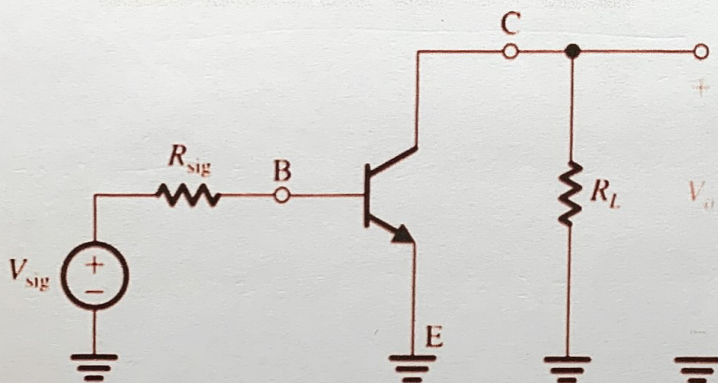
①
② 求得新的 $C_L^* = 101 \text{ fF}$

10.9 For the amplifier in Example 10.3, find the value of R_L that reduces the midband gain to half the value found. What value of f_H results? Note the trade-off between gain and bandwidth.

Ans. 2.44 k Ω ; 923 kHz

Example 10.3

Find the midband gain and the upper 3-dB frequency of the common-emitter amplifier of Fig. 10.13(a) for the following case: $I_E = 1$ mA, $R_{sig} = 5$ k Ω , $R_L = 5$ k Ω , $\beta_0 = 100$, $V_A = 100$ V, $C_\mu = 1$ pF, and $f_T = 800$ MHz. Also, determine the frequency of the transmission zero.



$$g_m = \frac{I_C}{V_T} = \frac{1}{25} \text{ A/V} ; r_\pi = \frac{\beta}{g_m} = 2.5 \text{ k}\Omega ; r_o = \frac{V_A}{I_C} = 100 \text{ k}\Omega ;$$

$$C_\pi + C_\mu = \frac{g_m}{\omega_T} = 8 \text{ pF} , C_\mu = 1 \text{ pF} \Rightarrow C_\pi = 7 \text{ pF}$$

$$R_L' = R_L \parallel r_o = 4.76 \text{ k}\Omega$$

$$\text{求新的 } R_L^* \text{, 使得 } R_L' \text{ 减半, } R_L^* \parallel r_o = 2.38 \text{ k}\Omega \Rightarrow R_L^* = 2.44 \text{ k}\Omega$$

$$C_{in}^* = C_\pi + C_\mu (1 + g_m \cdot 2.38 \text{ k}\Omega) = 103.20 \text{ pF}$$

$$f_H^* = \frac{1}{2\pi C_{in}^* \cdot R_{sig}} = 923.94 \text{ kHz}$$

10.14 Consider a bipolar active-loaded CE amplifier having the load current source implemented with a *pnp* transistor. Let the circuit be operating at a 1-mA bias current. The transistors are specified as follows: $\beta(npn) = 200$, $V_{An} = 130$ V, $|V_{Ap}| = 50$ V, $C_\pi = 16$ pF, $C_\mu = 0.3$ pF, and $C_L = 5$ pF. The amplifier is fed with a signal source having a resistance of 36 k Ω . Determine: (a) A_M ; (b) C_{in} and f_H using the Miller effect; (c) f_H using open-circuit time constants; (d) f_z ; and (e) the gain-bandwidth product.

Ans. (a) -176 V/V; (b) 450 pF, 80.6 kHz; (c) 73.5 kHz; (d) 21.2 GHz; (e) 12.9 MHz

$$g_m = \frac{I_C}{V_T} = \frac{1}{25} \text{ A/V}; \quad r_{on} = \frac{V_{An}}{I_C} = 130 \text{ k}\Omega; \quad r_{op} = \frac{|V_{Ap}|}{I_C} = 50 \text{ k}\Omega; \quad r_a = \frac{\beta}{g_m} = 5 \text{ k}\Omega$$

$$A_M = \frac{5}{36+5} \cdot [-g_m \cdot (r_{on} \parallel r_{op})] = -176 \text{ V/V}$$

$$C_{in} = C_a + [1 + g_m(r_{on} \parallel r_{op})] \cdot C_\mu = 449.632 \text{ pF}$$

$$(b) f_H = \frac{1}{2\pi(R_{sig} \parallel r_a) \cdot C_{in}} = 80.671 \text{ kHz} \quad (\text{using the Miller effect, 把 } C_{in} \text{ 关联的极点频率当作 } f_H)$$

$$(c) f_H = \frac{1}{2\pi(R_{sig} \parallel r_a) \cdot C_{in} + 2\pi \cdot r_{on} \parallel r_{op} \cdot (C_\mu + C_L)} = 73.54 \text{ kHz} \quad (\text{先 Miller 等效, 再把输入、输出极点都考虑进去, 用 open-circuit time constants 方法})$$

$$(d) f_z = \frac{g_m}{2\pi C_\mu} = 21.2 \text{ GHz} \quad (\text{参见教材 P.699})$$

$$(e) \text{Gain} \times \text{BW} = 176 \cdot 73.54 \text{ kHz} = 12.9 \text{ MHz}$$